MATH 11 — ELEMENTARY STATISTICS

§2.1

Frequency Distributions

Turning a pile of numbers into a table you can read

Today's Goal

- Create a frequency distribution table from raw data
- Construct grouped frequency distributions with appropriate class widths
- Calculate relative frequencies and cumulative frequencies
- Interpret frequency distributions to understand data patterns

What You Will Be Graded On

LT.5 Graphical Summaries of Data

I understand the different types of graphical displays for data (e.g., bar graphs, histograms, boxplots, etc.). I can interpret and create appropriate graphical summaries for various types of data.

A Pile of Answers

Twenty-five students were asked how many days last week they exercised.

4 0 3 7 2 5 4 1 6 3

4 7 2 5 3 6 1 4 7 2

5 3 6 4 7

Most common answer?

Counting Is the Whole Idea

Definition 2.1.1

The **frequency** of a particular data value refers to the number of times that value appears in the data set. A **frequency distribution** is a summary of how often different values occur in a data set.

This works for words as well as numbers. Eye color has a frequency distribution too.

One Column Per Answer

Example 1.

Build a frequency table for the twenty-five exercise answers.

Days exercised	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Frequency	1	2	3	4	5	3	3	4

The counts add to 25, so nobody was dropped. Four days is the most common answer.

Now Forty Bus Rides

Forty students said how many minutes their bus ride to campus takes.

34	18	47	61	28	12	39	50	25	72
44	31	56	20	36	82	30	45	15	38
64	26	51	33	21	48	68	35	76	28
40	23	37	58	54	32	24	52	42	39

One row for each value?

Put the Values in Bands

Example 2.

Summarize the forty bus times in a **grouped frequency table**.

Time (minutes)	**10–24**	**25–39**	**40–54**	**55–69**	**70–84**
Frequency	7	15	10	5	3

Thirty-eight different values became five bands, and the story is now visible: most rides are short.

The Parts of a Grouped Table

- The five **classes** are "10–24", "25–39", "40–54", "55–69", and "70–84".
- The **lower class limits** are $10, 25, 40, 55$, and 70.
- The **upper class limits** are $24, 39, 54, 69$, and 84.
- The **class width** is $25 - 10 = 15$.
- The **class midpoints** are $17, 32, 47, 62$, and 77.
- The **class boundaries** are $9.5, 24.5, 39.5, 54.5, 69.5$, and 84.5.

Six Steps

1. **Select the number of classes.** Usually between 5 and 20.
2. **Determine the class width.**
3. **Determine the first lower class limit.**
4. **Determine all other lower class limits.**
5. **Determine the upper class limits.**
6. **Determine the frequency for each class.**

Steps 1 to 3 on the Bus Times

1. **Number of classes.** Only 40 values, so choose 5.
2. **Class width.**

$$\frac{(\text{max}) - (\text{min})}{\text{number of classes}} = \frac{82 - 12}{5} = 14 \quad \longrightarrow \quad \text{round up to } 15$$

3. **First lower class limit.** The smallest value is 12. Pick 10: just below it, and friendly with a width of 15.

Steps 4 to 6 on the Bus Times

4. **Other lower class limits.** Add the width each time: $10, 25, 40, 55, 70$.
5. **Upper class limits.** The data are whole numbers, so add 14, one less than the width: $24, 39, 54, 69, 84$.
6. **Frequencies.** Count the values in each class: $7, 15, 10, 5, 3$.

$7 + 15 + 10 + 5 + 3 = 40$. Every ride is in exactly one class.

Counts Become Proportions

Definition 2.1.4

The **relative frequency** of a particular data value, x, is the ratio (expressed as a fraction, decimal, or percentage) of the frequency for that data value to the total number of data values. In mathematical notation, this is expressed as $RF(x)$.

The Relative Frequency Column

Time (minutes)	Frequency	Relative Frequency
10–24	7	$\frac{7}{40} = 0.175 = 17.5\%$
25–39	15	$\frac{15}{40} = 0.375 = 37.5\%$
40–54	10	$\frac{10}{40} = 0.25 = 25\%$
55–69	5	$\frac{5}{40} = 0.125 = 12.5\%$
70–84	3	$\frac{3}{40} = 0.075 = 7.5\%$

Proportions That Pile Up

Definition 2.1.6

The **cumulative relative frequency** of a particular data value, x, is the ratio of the sum of all the frequencies for data values less than or equal to x to the total number of data values. In mathematical notation, this is expressed as

$$CRF(x) = \sum_{i \leq x} RF(i).$$

Adding the Proportions as You Go

Time (minutes)	Relative Frequency	Cumulative Relative Frequency
10–24	17.5%	$\frac{7}{40} = 17.5\%$
25–39	37.5%	$\frac{22}{40} = 55\%$
40–54	25%	$\frac{32}{40} = 80\%$
55–69	12.5%	$\frac{37}{40} = 92.5\%$
70–84	7.5%	$\frac{40}{40} = 100\%$

Now You Try: Build All Three Columns

You Try 2.1.1. Twenty households were asked how many cars they own. Build the frequency, relative frequency, and cumulative relative frequency columns.

1	2	0	3	1	2	1	2	3	1
0	2	1	3	2	1	2	1	2	1

You Try 2.1.1: Checking Your Answers

Cars	Frequency	Relative Frequency	Cumulative Relative Frequency
0	2	$\frac{2}{20} = 10\%$	$\frac{2}{20} = 10\%$
1	8	$\frac{8}{20} = 40\%$	$\frac{10}{20} = 50\%$
2	7	$\frac{7}{20} = 35\%$	$\frac{17}{20} = 85\%$
3	3	$\frac{3}{20} = 15\%$	$\frac{20}{20} = 100\%$

The proportions add to 100%, and the last cumulative entry is 100%. Both are your check.